

Calvin Joyce

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Objective

Medical robotics R&D / New Product Development internship, **available Jan–Sep 2027**. Graduate researcher under **Prof. Michael C. Yip** (UCSD ARCLab); *Nature* co-author and 5× first author (ICRA, ISRR). Focusing on autonomy, controls, and learning for patient-contact robots, grounded in years of building the hardware myself, deploying on and improving real systems.

Education

University of California, San Diego · B.S. & M.S. Mechanical Engineering, GPA 3.8/4.0

B.S. 2024 · M.S. expected June 2027

Relevant coursework: Robotics, Haptics, Soft Robotics, Linear Control, Mechanical Design, Experimental Techniques

Skills

- **Medical Robotics & pHRI**: haptics & bilateral teleoperation, catheter/continuum robots, clinical safety thresholds, user studies, OptiTrack
- **Software & Control**: Python, C++, MATLAB, PyTorch, ROS 2; imitation learning (ACT), PPO, MPPI; NVIDIA Warp/Newton
- **Hardware**: SolidWorks + FEA, Fusion 360, DFMA, resin/FDM printing, laser cutting; design requirements & acceptance criteria, verification testing, test fixtures, capstan drives, motor drivers, sensors

Research & Work Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego · Advisor: Prof. Michael C. Yip

June 2021 – Present

Grad. Student Researcher (funded, full-time, 2025–); R&D Engineer (full-time, 2024–25); Research Asst. (paid, 2022–24)

La Jolla, CA

• PHABS (Portable Haptic Assisted Bimanual System)

Project lead; first author, ICRA 2027 submission

- * Invented the first handheld bimanual device to render the internal force between two hands on a shared object, the signal that decides crushing or dropping it, by reacting that force across the operator's hands so the device needs no grounding.
- * Built the bilateral teleoperation framework that renders a dual-arm follower's contact forces as per-hand pinch and signed inter-hand force, with both channels verified against a reference load cell.
- * Made demonstrations safer in a 10-person, 247-trial user study: pinch feedback cut egg crushes from 33% to 6% and grip force by two-thirds, and the inter-hand channel nearly halved squeeze on a shared-object lift.

• RIC (Robotic Infant Care) · clinical safety benchmark for robotic infant handling

Project lead; first author, ICRA 2027 submission

- * Introduced CRIB, the first framework scoring robots against clinical safety limits for handling infants, applied to bimanual pickup and CPAP nasal-mask repositioning, two of the NICU's most frequent hands-on tasks.
- * Converted clinical practice into testable acceptance criteria with four NICU clinicians: cervical pitch and head acceleration for pickup; per-site contact force, eye/mouth exclusion zones, and a 15 s PEEP budget for CPAP.
- * Benchmarked direct-human, teleoperated, and learned (ACT) handling across 641 instrumented trials on a Unitree G1 and an OpenArm, by building a motion-captured infant manikin and a force-sensing CPAP mask.
- * Showed force regulation is the gap: pickup success ran from 100% (human) to 83% (policy), but on CPAP humans met the pressure bound on just 1.7% of trials, against 61% for expert teleoperation.

• Humanoid Surgery (video) · in vivo feasibility of humanoid robots in laparoscopic surgery

Co-author, *Nature* 2026

- * Enabled humanoid robots to operate standard medical tools in dry-lab and in vivo porcine surgeries, by designing the tool-mount system (optimized mass/CoM, bench- and sim-validated rigidity) and running mechanical setup for every surgical session.
- * Coordinated a 10-person robot-clinician team through live surgeries, turning surgeon feedback into hardware revisions.

• Autonomous Catheter Navigation · continuum robotics; ARPA-H & DARPA project

Jul 2026 – Present

- * Building the learned low-level controller that drives a catheter tip to planner-issued waypoints, by training PPO policies in GPU physics on real patient anatomy, with contact limits anchored to measured tissue perforation forces.
- * Found and fixed a physics fault that had silently disabled the catheter's elasticity, by ruling out device, anatomy, drive, and stiffness with data before tracing it to our containment layer.

• Mentorship: 16+ students across lab projects, through CAD/FEA design reviews, DFMA, fixture design, and test plans.

ArtemisAI Labs · Mechanical Engineer, La Jolla, CA

Jan 2026 – Present

- Redesigned a NICU bedside camera mount to clinically deployable requirements (IEC 60601-1 instability test, no-disassembly wipe-down), now fielding the cameras for ArtemisAI's Phase II trial at Mount Sinai.

Triton Robotics, UC San Diego · Team President / Advisor

June 2020 – June 2023

- Scaled the team 20→80+ members and diversity 15%→35%+, by formalizing onboarding, DEI partnerships, and targeted recruiting; ran a 17-lead org with OKRs and roadmaps that won national nonprofit status, \$3K in funding, and permanent lab space.

Selected Research Projects

• Haptic Shoulder (paper) · pHRI phantom; co-first author, ICRA 2025; provisional patent

Jan 2024 – June 2024

Invented a biomechanically accurate shoulder phantom for safe pHRI testing, by dedicating one motor-encoder module to each anatomical DOF to reproduce joint center, coupled kinematics, and range of motion.

• ARCSnake V2 (paper) · amphibious screw-propelled snake robot, w/ NASA JPL; IROS 2026

June 2021 – Sep 2025

Increased screw output torque 40% by raising the torque ratio, aligning the belt drive, and cutting friction (shorter drive stack, relieved rubbing surfaces); engineered the mobile testbed that characterized screw performance across media (co-first author, ICRA 2023); designed NASU, the first Archimedes screw with reconfigurable angle of attack (first author, ISRR 2024).

Publications (Google Scholar)

• *Nature* 2026 (co-author): “In vivo feasibility study of humanoid robots in surgery”

• ICRA 2027 submissions (first author): “PHABS (Portable Haptic Assisted Bimanual System) for Force-Annotated Bimanual Shared Object Teleoperated Data”; “How Safe Is Robotic Infant Handling? Benchmarking Learning from Demonstration Against Clinical Thresholds”

• ICRA 2025 (co-first author): “Haptic Shoulder for Rendering Biomechanically Accurate Joint Limits for Human-Robot Physical Interactions”

• ISRR 2024 (first author): “NASU — Novel Actuating Screw Unit: Origami-Inspired Screw-Based Propulsion on Mobile Ground Robots”

• ICRA 2023 (co-first author): “Mobility Analysis of Screw-Based Locomotion and Propulsion in Various Media”

• Additional: “Humanoids in Hospitals” (arXiv 2025, co-2nd author); “LapSurgie” (ICRA 2026); “ARCSnake V2” (IROS 2026)